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SUPERSEDING
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MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE TTRS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductor, uncoated. Navy Standard size 1(7).
Second - Polyethylene insulation, type II, class L, grade 3 or 4 of L-P-390. Nominal wall thickness 0.017 inch; wall thickness may vary from nominal, as necessary, in order for completed cable to meet the specified electrical requirements. One black and one white or natural conductor for each pair.
Third - Insulation jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal wall thickness 0.003 inch.
Fourth - Two conductors (one black and one white or natural), cabled together to form a pair. The length of lay shall be such as to enable the completed cable to meet the twisting and bending endurance tests.
Fifth - Braided shield of AWG No. 34 or No. 36 copper uncoated or tin coated. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.
Sixth - Shield insulation of one polyester tape, type G of MIL-I-631, plus a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal wall thickness of 0.003 inch. Standard Identification Code, applied by Method 2. Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed. Standard Identification Code, applied by Method 2 on inner tape.
Seventh - The required number of pairs (see Details) cabled together with left-hand lay. Length of lay shall be such as to enable the completed cable to meet the twisting and bending endurance tests. Cabling sequence shall be consecutive, starting with Pair No. 1 from center outward. A single pair shall not be used as the central core of the cable. Fibrous fillers shall be used in the center and valleys, as necessary, to form a firm, round assembly.
Eighth - Binder tape applied helically with overlap.
Ninth - Standard thermoplastic jacket (Gray). Cable surface marking required.

Details

Type and size	Number of shielded pairs	Overall diameter maximum (inches)
TTRS-2	2	0.680
TTRS-4	4	.740
TTRS-6	6	.880
TTRS-8	8	.990
TTRS-10	10	1.080
TTRS-12	12	1.100
TTRS-16	16	1.190

Denotes changes

FSC 6145

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EXAMINATION AND TESTS

	Requirements:
Basic electrical:	
Conductor resistance - Ohms/1000 ft., at 25°C., maximum	10.3
Voltage withstand - Volts, rms, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	200
Group A:	
Visual and dimensional	
Mutual capacitance - pF/ft. at 1 MHz, maximum	28
Capacitance unbalance - Percent, maximum	10
Characteristic impedance - Ohms, at 1 MHz minimum	75
maximum	90
Group B:	
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	300
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	225
(C) Twisting and bending endurance - Note 1	
At 75° ± 2°C., cycles, minimum	6000
At minus 30° ± 2°C., cycles, minimum	2000

Note 1: Roller diameters for bending endurance test shall be as follows:

Cable type and size	Roller diameter (inches, maximum)
TTRS-2	2.0
TTRS-4	2.2
TTRS-6	3.0
TTRS-8	3.5
TTRS-10	4.0
TTRS-12	4.0
TTRS-16	4.5

Group C:	
Flammability - inches, maximum	2
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - Percent of unaged, minimum	60
Jacket (cable)	
Hot oil immersion	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	50
Permanence of printing (jacket) - Cycles, minimum	250

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical, plus all of Groups A, B, and C.

UNIT ORDERING LENGTHS:

All sizes 1500 feet (nominal)

(C) Custodians:
Army - MI
Navy - SH
Air Force - 80

(C) Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

(C) User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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